Weilin Li

Last updated: October 2026

Contact Information

Office: Room 308B, Marshak Science Building
City College of New York
160 Convent Ave, New York, NY 10031

Email: wli6@ccny.cuny.edu
Website: https://weilinlimath.github.io/
Google Scholar Profile

Research Interests

Applied analysis, computational mathematics, signal processing, machine learning

Employment

City University of New York
Doctoral Faculty, Mathematics Program, Graduate Center 2025–
Assistant Professor, Department of Mathematics, City College of New York 2022–

New York University
Visiting Assistant Professor 2021–22
Courant Instructor 2018–21

Education

University of Maryland, College Park 2018
Ph.D. Mathematics
Thesis: Topics in Harmonic Analysis, Sparse Representations, and Data Analysis
Advisors: John J. Benedetto and Wojciech Czaja

Cornell University 2013
B.A. Mathematics, Summa Cum Laude
Thesis: Boundary Value Problems on a Half-Sierpinski Gasket
Advisor: Robert S. Strichartz

Preprints

3. Ernesto Araya, Gitta Kutyniok, Weilin Li, Johannes Maly, and Manjot Singh. Formalizing the intrinsic dimension of objective landscapes by a generalized notion of co-dimension. [arXiv]
2. Albert Fannjiang and Weilin Li. Multidimensional Gradient-MUSIC: A global nonconvex optimization framework for optimal resolution. [arXiv]
1. C. Sinan Güntürk and Weilin Li. Approximation of functions with one-bit neural networks. [arXiv]

Journal Publications

21. Hana Huber and Weilin Li. A sharp analysis of Root-MUSIC: locations of correct and extraneous roots. *Sampling Theory, Signal Processing, and Data Analysis*, to appear. [arXiv]
20. Albert Fannjiang and Weilin Li. Optimal structured approximation of Fourier subspaces, Toeplitz matrices, and exponential sums. *Linear Algebra and its Applications*, vol. 752, pp. 372–421, 2027. [journal]
19. Albert Fannjiang, Weilin Li, and Wenjing Liao. Optimality of Gradient-MUSIC for spectral estimation. *Information and Inference*, vol. 15, no. 3, 2026. [journal]

18. Weilin Li, Karl Otness, Kui Ren, and Donsub Rim. An explicit spectral decomposition of the ADRT. *BIT Numerical Mathematics*, vol. 66, no. 37, 2026. [journal]
17. Oleg Asipchuk, Laura De Carli, Weilin Li. Concerning the stability of exponential systems and Fourier matrices. *Journal of Fourier Analysis and Applications*, vol. 31, no. 67, 2025. [journal]
16. Weilin Li. Nonharmonic multivariate Fourier transforms and matrices: condition numbers and hyperplane geometry. *Applied and Computational Harmonic Analysis*, vol. 79, pp. 101791, 2025. [journal]
15. Sjoerd Dirksen, Weilin Li, and Johannes Maly. Subspace and DOA estimation under coarse quantization. *IEEE Transactions on Information Theory*, vol. 71, no. 10, pp. 8149–8168, 2025. [journal]
14. Weilin Li. Multiscale estimates for the condition number of non-harmonic Fourier matrices. *Mathematics of Computation*, vol. 94, no. 356, pp. 2895–2929, 2025. [journal]
13. C. Sinan Güntürk and Weilin Li. Approximation with one-bit polynomials in Bernstein form. *Constructive Approximation (special issue for Ron DeVore's 80th birthday)*, vol. 57, no. 2, pp. 601–630, 2023. [journal]
12. Weilin Li, Kui Ren, and Donsub Rim. A range characterization of the single-quadrant ADRT. *Mathematics of Computation*, vol. 92, no. 339, pp. 283–306, 2023. [journal]
11. C. Sinan Güntürk and Weilin Li. Quantization for spectral super-resolution. *Constructive Approximation*, vol. 56, no. 3, pp. 619–648, 2022. [journal]
10. Weilin Li, Zengying Zhu, Weiguo Gao, and Wenjing Liao. Stability and super-resolution of MUSIC and ESPRIT for multi-snapshot spectral estimation. *IEEE Transactions on Signal Processing*, vol. 70, pp. 4555–4570, 2022. [journal]
9. Ilya Kavalerov, Weilin Li, Wojciech Czaja, and Rama Chellappa. 3-D Fourier scattering transform and classification of hyperspectral images. *IEEE Transactions on Geoscience and Remote Sensing*, vol. 59, no. 12, pp. 10312–10327, 2021. [journal]
8. Weilin Li and Wenjing Liao. Stable super-resolution limit and smallest singular value of restricted Fourier matrices. *Applied and Computational Harmonic Analysis*, vol. 51, no. 1, pp. 118–156, 2021. [journal]
7. Weilin Li. Generalization error of minimum weighted norm and kernel interpolation. *SIAM Journal on Mathematics of Data Science*, vol. 3, no. 1, pp. 414–438, 2021. [journal]
6. Weilin Li, Wenjing Liao, and Albert Fannjiang. Super-resolution limit of the ESPRIT algorithm. *IEEE Transactions on Information Theory*, vol. 66, no. 7, pp. 4593–4608, 2020. [journal]
5. Wojciech Czaja and Weilin Li. Rotationally invariant time-frequency scattering transforms. *Journal of Fourier Analysis and Applications*, vol. 26, no. 1, pp. 1–23, 2020. [journal]
4. John J. Benedetto and Weilin Li. Super-resolution by means of Beurling minimal extrapolation. *Applied and Computational Harmonic Analysis*, vol. 48, no. 1, pp. 218–241, 2020. [journal]
3. Wojciech Czaja and Weilin Li. Analysis of time-frequency scattering transforms. *Applied and Computational Harmonic Analysis*, vol. 47, no. 1, pp. 149–171, 2019. [journal]
2. John Peterson, Weilin Li, Brian Cesar-Tondreau, John Bird, Kevin Kochersberger, Wojciech Czaja, and Morgan McLean. Experiments in unmanned aerial vehicle/unmanned ground vehicle radiation search. *Journal of Field Robotics*, vol. 36, no. 4, pp. 818–845, 2019. [journal]
1. Weilin Li and Robert S. Strichartz. Boundary value problems on a half Sierpinski gasket. *Journal of Fractal Geometry*, vol. 1, no. 1, pp. 1–43, 2014. [journal]

Refereed Book Chapters

1. Wojciech Czaja, Weilin Li, Yiran Li, and Mike Pekala. Maximal function pooling with applications. *Excursions in Harmonic Analysis, Volume 6 (special volume for John Benedetto's 80th birthday)*, pp. 413–429, 2021. [book chapter].

Refereed Conference Proceedings

9. Weilin Li. New perspectives on Fourier matrices. *15th International Conference on Sampling Theory and Applications (SampTA)*, 2025. [proceeding]
8. Sjoerd Dirksen, Weilin Li, and Johannes Maly. Subspace estimation under coarse quantization. *15th International Conference on Sampling Theory and Applications (SampTA)*, 2025. [proceeding]
7. C. Sinan Güntürk and Weilin Li. Uniform approximation by polynomials with integer coefficients via the Bernstein lattice. *Combinatorial and Additive Number Theory, New York Number Theory Seminar*, vol. 6, pp. 221–231, 2025. [proceeding]
6. C. Sinan Güntürk and Weilin Li. Near-optimality of $\Sigma\Delta$ quantization for L^2 approximation with polynomials in Bernstein form. *14th International Conference on Sampling Theory and Applications (SampTA)*, 2023. [proceeding]
5. Wojciech Czaja, Ilya Kavalerov, and Weilin Li. Exploring the high dimensional geometry of HSI features. *11th Workshop on Hyperspectral Imaging and Signal Processing: Evolution in Remote Sensing (WHISPERS)*, pp. 1–5, 2021. [proceeding]
4. Weilin Li and Wenjing Liao. Conditioning of restricted Fourier matrices and super-resolution of MUSIC. *13th International Conference on Sampling Theory and Applications (SampTA)*, 2019. [proceeding]
3. C. Sinan Güntürk and Weilin Li. High performance quantization for spectral super-resolution. *13th International Conference on Sampling Theory and Applications (SampTA)*, 2019. [proceeding]
2. Kevin Kochersberger, John Peterson, Prashant Kumar, John Bird, Morgan McLean, Wojciech Czaja, Weilin Li, and Nathaniel Monson. Unmanned aircraft applications in radiological surveys. *IEEE International Symposium on Technologies for Homeland Security*, pp. 1–5, 2018. [proceeding]
1. Wojciech Czaja, Ilya Kavalerov, and Weilin Li. Scattering transforms and classification of hyperspectral images. *SPIE, Algorithms and Technologies for Multispectral, Hyperspectral, and Ultraspectral Imagery*, XXIV, 2018. [proceeding]

External Grants

3. *National Science Foundation, Division of Mathematical Sciences, Sole-PI Grant* 09/2026–
A sole-PI project "A rigorous optimization framework for high-resolution parameter estimation on the continuum" awarded by the NSF Computational Mathematics program with a budget of $215,000. Award number # 2607872.
2. *National Science Foundation, Division of Mathematical Sciences, Sole-PI Grant* 09/2023–09/26
A sole-PI project "Multidimensional and compressive super-resolution: theory, computation, and fundamental limits" awarded by the NSF Computational Mathematics program with a budget of $176,476. Award number # 2309602.
1. *AMS-Simons Travel Grant* 2019–22
Awarded $5,000 for travel, funded by the AMS-Simons Travel Grants Program.

Internal Grants

5. *PSC-CUNY Track I Grant* 2026–
A sole-PI project "An optimization framework for deconvolution" with a budget of $6,000.
4. *PSC-CUNY Traditional B Grant* 2025–26
A sole-PI project "Kernel matrices, extremal functions, and applications" with a budget of $6,000.
3. *PSC-CUNY Traditional B Grant* 2024–25
A sole-PI project "A proof that ESPRIT is optimal" with a budget of $4,500.

2. *PSC-CUNY Traditional B Grant* 2023–24
A sole-PI project "Multidimensional super-resolution algorithms" with a budget of $5,450.

1. *Cohen Foundation Research Support, New York University* 2018–19
Awarded $5,000 for travel and research equipment, funded by the Cohen Foundation.

Awards and Honors

5. *Charles Chui Young Researcher Best Paper Award* 2021
for "Stable super-resolution limit and smallest singular value of restricted Fourier matrices"

4. *James C. Alexander Prize for Graduate Research, University of Maryland* 2018
One of two PhD students who won this prize for outstanding thesis.

3. *Program Associate, Mathematical Sciences Research Institute* 2017
Participated as a student visitor for the 2017 Harmonic Analysis program at Mathematical Sciences Research Institute, Berkeley California.

2. *Ann G. Wylie Dissertation Fellowship, University of Maryland* 2017
Competitive university fellowship that was awarded to 50 graduate students in 2017.

1. *Graduated Summa Cum Laude in Mathematics, Cornell University* 2013
Graduated with highest honors in mathematics from Cornell University.

Recorded Online Talks

3. "A nonconvex optimization approach to spectral estimation" 2026
1W-MINDS: One World Mathematics of INformation, Data, and Signals

2. "Function approximation with one-bit Bernstein and neural networks" 2021
Faraway Fourier Talks

1. "Super-resolution, subspace methods, and Fourier matrices" 2021
1W-MINDS: One World Mathematics of INformation, Data, and Signals

Conference Talks

21. Mini-symposium on Machine learning theory for neural networks and scientific modeling 2026
SIAM New York-New Jersey-Pennsylvania Sectional Meeting

20. Oral Presentation 2025
15th International Conference on Sampling Theory and Applications, Vienna, Austria

19. Invited talk 2024
Frame Theory Days, St. Louis University, MO

18. Invited talk 2024
NSF PI Meeting, University of Washington Seattle, WA

17. Mini-symposium on Nonlinear Approximation and Numerical Linear Algebra 2024
SIAM Conference on Applied Linear Algebra, Paris, France

16. Special session on Bases and Frames in Hilbert Spaces 2024
AMS Spring Sectional Meeting, Tallahassee, FL

15. Session on Theory and Algorithms of Super-Resolution in Imaging and Inverse problems 2023
11th Applied Inverse Problems Conference, Göttingen, Germany

14. Inverse Problems and Reduced Order Modeling for Wave Propagation Problems 2023
17th US National Congress on Computational Mechanics, Albuquerque, New Mexico

13. Workshop on Mathematical Signal and Image Analysis 2023
TUM Science and Study Center Raitenhaslach, Germany

12. Fall Fourier Talks 2022
University of Maryland, College Park, MD
11. Focus Program on Data Science, Approximation Theory, and Harmonic Analysis 2022
Fields Institute, Toronto, Canada
10. AMS Special Session on Geometry in the Mathematics of Data Science 2022
Joint Math Meetings, Seattle, WA
9. Mini-symposium on Recovering Algebraic Structures in Imaging and Signal Processing 2022
SIAM Conference on Imaging Science, Berlin, Germany
8. 11th Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing 2021
Amsterdam, Netherlands (online)
7. Workshop on Mathematical Signal and Image Analysis 2020
Raitenhaslach, Germany (canceled due to covid)
6. Randomness and Determinism in Compressive Data Acquisition 2019
Texas A&M University, College Station, TX
5. Special Session on Applied Harmonic Analysis: Frame Theory and Applications 2018
AMS Fall Western Sectional Meeting, San Francisco State University, CA
4. 7th International Conference on Computational Harmonic Analysis 2018
Vanderbilt University, Nashville, TN
3. Special Session on Applied Harmonic Analysis: Frames, Samplings and Applications 2017
AMS Sectional Meeting, University of Florida, Orlando, FL
2. AMS Special Session on Fractal Geometry: Mathematics of Fractals and Related Topics 2014
Joint Mathematics Meetings, Baltimore, MD
1. AMS Session on Ergodic Theory, Dynamics, and Harmonic Analysis 2013
Joint Mathematics Meetings, San Diego, CA

Seminar Talks

29. Numerical Analysis Seminar, University of Maryland, College Park, MD 2026
28. Mathematics of Machine Learning, University of Pittsburgh, PA 2026
27. Harmonic Analysis and PDE seminar, CUNY Graduate Center, New York, NY 2025
26. Colloquium, New Jersey Institute of Technology, Newark, NJ 2025
25. Analysis and Data Science Seminar, SUNY Albany, NY 2024
24. Data-Enabled Science Seminar, University of Houston, TX 2024
23. Applied and Computational Math Seminar, Georgia Institute of Technology, Atlanta, GA 2024
22. Colloquium, Florida International University, Miami, FL 2023
21. Machine Learning Seminar, University of Massachusetts at Amherst, MA 2023
20. Colloquium, The City College of New York, New York, NY 2022
19. Applied Math Colloquium, Columbia University, New York, NY 2022
18. Harmonic Analysis and PDE Seminar, CUNY Graduate Center (online) 2022
17. Applied/PDE/Data Seminar, University of California, Santa Barbra, CA (online) 2021
16. Scientific Computing And Numerics, Cornell University, Ithaca, NY (online) 2021
15. Computational and Applied Math Seminar, Tufts University, Medford, MA 2021
14. Computational Analysis Seminar, Vanderbilt University, Nashville, TN (online) 2021

13. Colloquium, Washington University in St. Louis, MO (online) 2021
12. Colloquium, North Carolina State University, Raleigh, NC (online) 2020
11. Seminar in Mathematical Physics/PDE, University of California, San Diego, CA 2019
10. Applied and Computational Math Seminar, University of California, Irvine, CA 2019
9. Norbert Wiener Center Seminar, University of Maryland, College Park 2019
8. Analysis Seminar, Courant Institute of Mathematical Sciences, New York, NY 2019
7. Norbert Wiener Center Seminar, University of Maryland, College Park, MD 2018
6. Applied Math and Analysis Seminar, Duke University, Raleigh, NC 2018
5. Harmonic Analysis and Signal Processing Seminar, Courant Institute, New York, NY 2017
4. Applied and Computational Math Seminar, Georgia Institute of Technology, Atlanta, GA 2017
3. Norbert Wiener Center Seminar, University of Maryland, College Park, MD 2017
2. Graduate Student Seminar, Mathematical Sciences Research Institute, Berkeley, CA 2017
1. Research Interaction Team on Deep Learning, University of Maryland, College Park, MD 2016

Teaching Experience

City College of New York: (* indicates a graduate level class)

- Math 434/A34* Theory of Functions of a Real Variables (Fa23, Sp25, Fa26)
- Math 432/A32* Theory of Functions of a Complex Variable (Fa25)
- Math 377 Applied Statistics and Probability (Sp23, Sp24)
- Math 328 Methods of Numerical Analysis (Fa24)
- Math 324 Advanced Calculus II (Fa22)
- Math 213 Calculus III with Planar Vector Analysis (Sp26)

New York University: (* indicates a graduate level class)

- Math 2110* Linear Algebra I (Fa21)
- Math 2111* Linear Algebra (Fa21)
- Math 325 Analysis (Sp22, Sp21, Sp20)
- Math 211 Math for Economics I (Fa19)
- Math 123 Calculus III (Sp19)
- Math 122 Calculus II (Fa20)
- Math 121 Calculus I (Fa18)